FINANCIAL TRANSACTION PATTERN ANALYSIS USING BIG DATA ANALYTICS

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ABSTRACT

Financial institutions generate a large amount of transaction data due to the rapid growth of digital payments and online banking. Analyzing this data is important to understand transaction patterns and identify suspicious activities. This project focuses on Financial Transaction Pattern Analysis using Cloud-Based Data Analytics.

The dataset used in this project is stored and analyzed in a cloud environment, which provides scalable and secure data storage. The data contains missing values, duplicates, and inconsistent formats. Data preprocessing and cleaning are performed to improve data quality. After cleaning, analytical methods are applied to analyze transaction patterns such as high-value transactions and customer behavior.

The cloud platform enables efficient storage, processing, and analysis of financial data. This project demonstrates how cloud-based data analytics helps extract useful insights from financial transaction data and supports better decision-making in financial systems.

EXISTING SYSTEM

- Financial transaction data is stored in traditional databases or local systems
- Basic SQL queries and spreadsheets are used for analysis
- Some systems use simple transaction monitoring tools
- Data is often stored in local servers instead of cloud platforms
- Manual analysis and verification is performed for suspicious transactions
- Limited use of cloud-based analytics platforms

DRAWBACKS OF EXISTING SOLUTIONS

- Local systems have limited storage capacity
- Difficult to handle large volumes of financial data efficiently
- Data access is slower and less flexible
- Manual analysis takes more time
- Difficult to analyze unclean and inconsistent data
- Lack of scalability compared to cloud platforms
- Difficult to identify suspicious transactions quickly

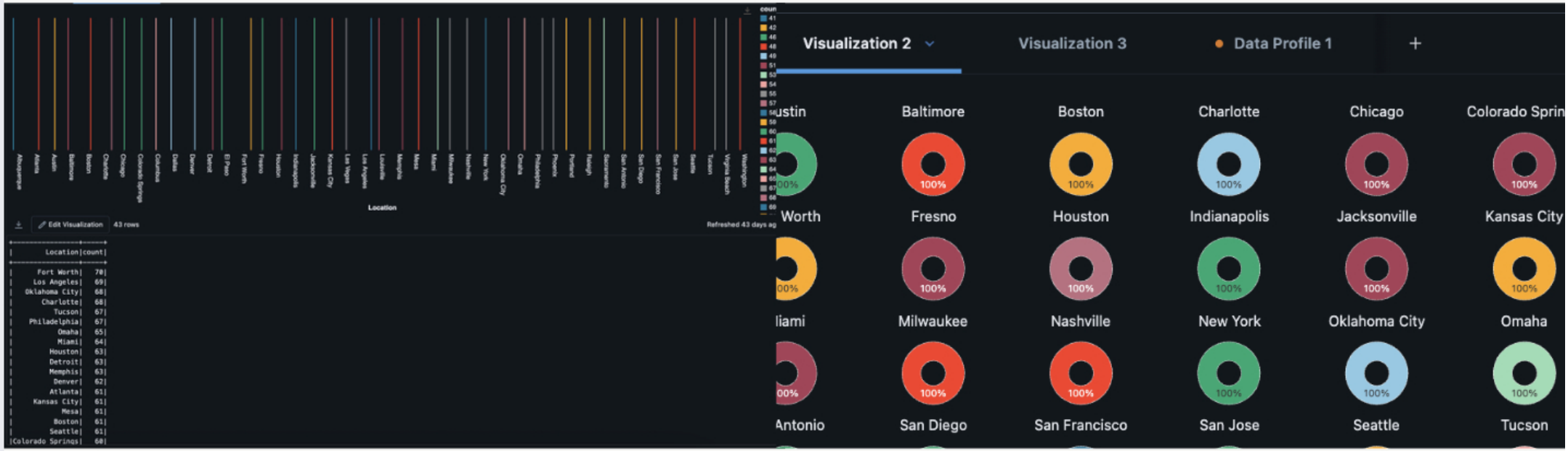
SOLUTION USED FOR THIS PROJECT

- Cloud Platform: Microsoft Azure
- Cloud Storage: Azure Databricks File System (DBFS) / Cloud storage
- Data Processing Tool: Azure Databricks / Google Colab (Cloud-based notebook)
- Programming Language: Python
- Data Analysis Library: Pandas
- Data Preprocessing: Handling missing values, removing duplicates
- Dataset Format: CSV (Financial transaction dataset)
- Analysis Method: Transaction pattern analysis using cloud environment

CONCLUSION

- Financial transaction data was analyzed using cloud-based analytics.
- Cloud storage enabled efficient data handling.
- Data preprocessing improved data quality.
- Transaction patterns were successfully identified.
- Cloud analytics helps in better financial data analysis

results



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Los Angeles	69
Oklahoma City	68
Charlotte	68
Tucson	67
Philadelphia	67
Omaha	65
Miami	64
Houston	63
Detroit	63
Memphis	63
Denver	62
Atlanta	61
Kansas City	61
Mesa	61
Boston	61
Seattle	61
Colorado Springs	60

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Table
Visualization 1
Visualization 2
Visualization 3
Data Profile 1
Albuquerque
Atlanta
Austin
Baltimore
Boston
Charlotte
Chicago
Colorado Springs
Columbus
Dallas
Denve.
Detroit
El Paso
Fort Worth
Fresno
Houston
Indianapolis
Jacksonville
Kansas City
Las Vegas
Los Angeles
Louisville
Memphis
Mesa
Miami
Milwaukee
Nashville
New York
Oklahoma City
Omaha
Philadelphia
Phoenix
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Raleigh
Sacramento
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San Diego
San Francisco
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